

[Technical Implementation and Reproducibility Details]

BrainAgent: Multi-Agent Fusion for Glioma Report Generation from Multi-Sequence MRI

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This document accompanies the IEEE Transactions on Medical Imaging submission of the same title. Cross-references of the form “Section II-G3” and “Fig. 3” point into the main paper; cross-references of the form “Appendix A-B” point within this document. Float numbering continues that of the main paper, which ends at Fig. 4 and Table I, so Figs. 5–9 and Tables II–IX below belong to this document. Numbered equations (1) and (2) are those of the main paper. It collects the implementation detail, the extended analyses and the reproducibility record that the page limit does not accommodate; no result reported in the main paper depends on content that is available only here.

APPENDIX A METHODS DETAILS

A. Extended Related Work: Text-Native Language Models in Neuroradiology

Recent advancements in large language models (LLMs) have made automated reporting plausible. LLMs now encode substantial clinical knowledge [1] and are increasingly woven into diagnostic and triage workflows across medicine [2]. In neuroradiology specifically, several rigorous studies establish both the promise and the boundaries of the text-native paradigm. A multicenter benchmark and reader study of ten LLMs on 4,293 brain MRI reports found that a frontier reasoning model (DeepSeek-R1) could generate diagnostic impressions at a level approaching radiologists, reaching 97.6% patient-level accuracy [3]. A systematic review and meta-analysis comparing LLMs against healthcare professionals showed competitive and complementary diagnostic and triage performance, and demonstrated that fusing diagnostic clinical text with imaging metadata improves accuracy [4]. A randomized experiment showed that chain-of-thought rationales improved physician diagnostic accuracy in radiology by 12.2% [5]. A large scoping review confirmed that LLM-based information extraction from radiology reports is dominated by MRI and CT, and that transformer models can convert unstructured radiology prose into structured clinical data [6]. Broad surveys have begun to chart this fast-growing landscape, mapping the proliferation of LLMs and multimodal LLMs across medicine [7] and, more generally, the deployment of LLMs throughout healthcare [8].

1) *Clinically oriented report-generation metrics:* The field’s earliest report-generation metrics capture surface n-gram overlap rather than clinical correctness; the response has been a succession of clinically oriented metrics CheXbert-F1 [9], RadGraph-F1 [10], BERTScore [11], the composite RadCliQ [12], and the LLM-based GREEN score [13] which largely developed and validated on chest radiography.

B. Provenance of the Pretrained Components

No agent in this pipeline was trained on the data it is tested on. The sixteen agents (twelve modality-view writers, three view-level consolidators and the master orchestrator) all run on a single vision-language backbone used exactly as released; that backbone was not fine-tuned, adapted or prompt-tuned on RHUH-GBM, UCSF-PDGM or the in-house cohort at any stage of this work, and the LLM judge is likewise a third-party model used at inference only. MedSAM2 / SAM2.1 Hiera-Tiny [14], [15] and the skull-stripping tools HD-BET, ANTs and SynthStrip [16]–[18] are published third-party checkpoints used as released; the MNI ICBM152 2009a template [19] and the Harvard-Oxford and Pauli atlases [20], [21] are population atlases that are not learned from any of the cohorts.

The ConvNextLocator detector is the one component trained by the authors, and it carries a training overlap that we state here explicitly rather than leave to be discovered. It was trained on the UCSF-PDGM collection, using the expert tumour annotations distributed with that collection to supply slice-level tumour-presence labels and bounding boxes, and the patients it was trained on are the same patients on which the UCSF-PDGM reporting results of Section III-D of the main paper are obtained. On that cohort the detector is therefore evaluated on data it was fitted to.

The consequence is specific and bounded. On every cohort the agents are instructed to trust the pipeline’s quantitative measurements rather than re-estimate them (Section IV of the main paper), so report content follows the detector’s output wherever the detector fires; that design is common to all three datasets and is not what distinguishes UCSF-PDGM. What distinguishes UCSF-PDGM is that those measurements come from a detector fitted to the very patients being scored, so the localization, the bounding-box prompt passed to MedSAM2, and the volumetry and atlas involvement derived from the resulting mask are all more accurate on that cohort than they would be on unseen data. The UCSF-PDGM composite of 82.1% should accordingly be read as an optimistic bound rather than as evidence that the segmentation front end generalizes. It does not affect the internal validity of the human rubric itself, which compares an AI report against the patient’s images and not against the detector.

RHUU-GBM and the in-house teleradiology cohort contributed no training slices to the detector, and neither was used to train, fine-tune or select any component of the system. Both are therefore fully external, and UCSF-PDGM is the only one of the three cohorts on which any part of the pipeline was fitted to the data it is scored on.

Each cohort accordingly carries its own distinct caveat, and they should not be conflated: the RHUU-GBM figures are obtained under a favourable reference standard (Section III-C of the main paper), the UCSF-PDGM figures under the detector overlap described here, and the in-house cohort is the only one of the three subject to neither.

C. Cohort Heterogeneity and the Module Cascade

In contrast to the two open-source collections, this dataset is deliberately heterogeneous in scanner manufacturer, field strength (ranging from 1.5 T to 3 T), in-plane resolution, and slice spacing, indicating the mixed acquisition conditions encountered in routine neuro-oncology workflows. The four sequences required by BrainAgent (T1, T1c, T2 and FLAIR) are available for every case in raw DICOM format, but no expert-refined segmentation masks are provided. Consequently, experiments on the in-house data rely exclusively on the masks produced by the automatic segmentation module.

As illustrated in Fig. 1, the BrainAgent pipeline processes each patient through ten sophisticated modules, where module outputs serve as inputs for downstream stages. Raw MRI data is ingested and processed for the remaining modules. The pipeline orchestrator discovers and groups files by patient identifier, then executes the following cascade for each patient: (1) input validation and optional DICOM-to-NIfTI conversion, required for all later modules; (1a) image preprocessing, which feeds into later segmentation; (1b) optional skull stripping, used for improved tumor segmentation; (2) tumor segmentation, required for metrics computation; (3) tumor metrics computation, which informs following modules; (4) MNI-152 registration [19], needed before atlas-based analysis; (5) atlas-based region analysis; (6–7) slice extraction and batch image assembly, which provide input for report generation; (8) multi-agent LLM report generation, dependent on assembled images; (9) final report consolidation; and (10) 3D surface mesh generation, utilizing segmentation outputs.

1) *Evaluator Assignment*: One dataset was obtained from a partner clinical organization whose clinical workflow connects participating healthcare providers with subspecialty radiologists who author the corresponding reports as part of routine clinical practice. Report pairs from this dataset were reviewed by two senior radiologists from the organization to determine the system’s applicability in a clinical setting. The other two datasets were sourced from publicly available benchmarks and were each reviewed by an independent researcher from a separate, unaffiliated institution. Assigning a different evaluator group to each dataset, and specifically selecting external evaluators for the open datasets, was intended to minimize single-site and single-rater bias and to facilitate cross-institutional validation of the pipeline’s output.

2) *Input Handling and DICOM Conversion*: The pipeline supports five different folder structures to handle varied clinical and research data sources, including DICOM-based formats from clinical PACS systems.

The metadata of the patient is usually extracted from the DICOM files if they are available, especially the age. DICOM images are converted to NIfTI.

D. Quality-Assurance Protocol for the RHUH-GBM Reference Reports

A trial view of how this evaluation is carried out is available at <https://www.mednextsolutions.com/trial>.

The RHUH-GBM collection is distributed without free-text reports, so a reference standard had to be created for this study. Because those reports were written for this study rather than taken from clinical records, the authoring procedure is itself part of the method and is specified here in full, including the respects in which it falls short of a formal double-reporting protocol. No automated drafting step of any kind was used: every reference report was written by a neuroradiologist from the images. The 40 reports are released with the article so that the reference standard can be inspected and re-used independently.

- 1) **Readers.** Each reference report was authored by a single senior board-certified neuroradiologist working from the patient’s own multi-sequence MRI. Authoring was single-reader throughout: no case was independently reported by a second reader, and no adjudication step was therefore required or performed.
- 2) **Information available to the reader.** Readers had access to the four structural sequences distributed with the collection (T1, T1c, T2 and FLAIR), the available clinical metadata, and the tumour sub-region volumes released with RHUH-GBM. The consequence of that last item for the interpretation of the results is taken up under limitations below.
- 3) **Reporting procedure.** Reports follow a consistent section-based structure, but no item-by-item checklist was imposed and no per-finding accept/correct/delete decision was recorded. Readers wrote free text within that structure, covering findings, diagnosis and management content as they judged appropriate for the case.
- 4) **Audit trail.** Each released reference report carries the identity of its author and the date of authoring. Intermediate versions and per-case edit histories were not retained, so the released artifact is the final report together with its provenance stamp, not a reconstruction of the writing process.
- 5) **Residual limitations.** Three limitations follow from this design and are stated here rather than left implicit. First, reporting was single-reader, so the reliability of the reference standard itself cannot be quantified: there is no inter-reader agreement figure for the RHUH-GBM ground truth, unlike the rubric evaluations reported in Sections III-B and III-D of the main paper. Second, because no fixed checklist was applied, the completeness of each report cannot be audited item by item; a finding the reader did not mention leaves no trace. Third, as stated in Section III-C of the main paper, the reference is template-consistent and was authored with knowledge of the tumour sub-region volumes, which plausibly favours agreement with a pipeline that reports the same quantities in the same order. For all three reasons the RHUH figures are reported as an upper bound obtained under a favourable reference standard, not as an estimate of clinical performance.

E. Imaging Pipeline Details

1) *RAS+ Reorientation:* All images are reoriented to the Right-Anterior-Superior (RAS+) canonical orientation through analyzing the NIfTI affine matrix and applying the necessary axis permutations and flips. This is the RAS+ convention used throughout the pipeline, consistent with the NIfTI standard and MNI-152 space. The X-axis encodes right (+) to left (−), the Y-axis encodes anterior (+) to posterior (−), and the Z-axis encodes superior (+) to inferior (−). This standardizing process is critical for three downstream operations: (i) hemisphere determination in tumor metrics, where centroid $X < 0$ indicates left hemisphere; (ii) MNI-152 registration, which assumes RAS+ input; and (iii) consistent laterality marker placement on visualization slices.

2) *N4 Bias Field Correction:* Intensity non-uniformity arising from RF coil inhomogeneity is corrected using the N4 algorithm. The correction operates at four multi-resolution levels with iteration numbers of [50, 50, 30, 20] and convergence tolerance of 10^{-7} , using a shrink factor of 4 and a B-spline knot spacing of 200 mm. Bias correction is performed before skull stripping because the additional tissue context provides a more complete basis for estimating the low-frequency bias field [22].

3) *Isotropic Resampling:* Volumes are resampled to 1 mm isotropic voxel spacing using linear interpolation for anatomical images. Segmentation masks are resampled with nearest-neighbor interpolation to preserve discrete label values. A minimum volume threshold of 1,000 voxels is enforced to reject corrupt or truncated input files.

4) *Volumetry:* Total and per-class tumor volumes are computed from the number of voxels with label k and the voxel dimensions in millimeters obtained from the image header.

5) *Centroid and Hemisphere Determination:* The tumor centroid is computed in world coordinates using the NIfTI affine transformation matrix. Hemisphere determination uses a 5 mm threshold around the midline: $c_x < -5$ mm indicates left hemisphere, $c_x > +5$ mm indicates right hemisphere, and $|c_x| \leq 5$ mm indicates mid-line/bilateral involvement.

6) *Registration Transforms and Interpolation:* The registration is performed on a reference structural image and produces two transform components: a non-linear warp field and a linear affine matrix. These transforms are then applied to all remaining modality volumes and the segmentation mask. Linear interpolation is used for anatomical images; nearest-neighbor interpolation is used for segmentation masks to preserve discrete label values.

The complete numeric settings for detection, mask refinement, preprocessing, skull stripping, registration and tumour metrics are tabulated in Appendix C-G.

F. System Engineering Details

1) *Model-Agnostic Architecture:* It is important to note that our pipeline is highly dynamic and can be used with any vision-language model. As previously mentioned, the VLM is just a backbone for our agentic sub-system that can be replaced with any other model. By maintaining a model-agnostic architecture, the framework facilitates seamless integration with both future and past open- and closed-source models.

2) *View-Level Consolidation: Worked Example:* For example, the Axial consolidation agent receives the reports from $a_{T1,AX}$, $a_{T1c,AX}$, $a_{T2,AX}$, and $a_{FLAIR,AX}$ along with the full patient context.

3) *Structured Output Validation:* All LLM outputs are parsed into validated structured data models through a multi-strategy extraction pipeline designed to handle the variability of real-world LLM outputs. The extraction sequence runs as follows: (i) any chain-of-thought preamble or thinking tags are stripped; (ii) the system searches for code-fenced JSON blocks; (iii) a state-machine balanced-brace scanner identifies all well-formed JSON objects, correctly handling nested braces and quoted strings with escape characters; (iv) candidates are de-duplicated and the longest valid JSON object is selected after validation. Parsing failures are logged with full context, and the raw LLM response is always preserved for post-hoc analysis.

The structured output schema for per-modality reports includes fields for modality name, acquisition technique, a list of objective findings, numbered clinical impressions, follow-up recommendations, study limitations, and critical/urgent findings. The final consolidated report schema comprises two narrative sections: comprehensive findings and clinical impression with recommendations.

4) *Centralized Configuration:* All pipeline parameters are managed through a validated settings model that loads values from environment variables with type-safe defaults. The configuration encompasses more than 30 parameters spanning: preprocessing toggles (bias correction, resampling, target spacing); segmentation model paths, confidence thresholds, primary modality selection, and multi-plane/ MedSAM2 enablement; skull stripping method selection (as we have more than one method for skull-stripping); LLM inference mode, model names, server URLs, and maximum concurrent workers; and batch processing parameters (grid size, columns).

5) *Performance Monitoring:* A dedicated timing tracker records execution duration for each pipeline step and each LLM agent call using monotonic, high-resolution timers. The tracker generates both human-readable timing reports per-module percentage breakdowns, per-agent subtotals grouped by modality and view, and total LLM time fraction and machine-readable structured reports.

6) *Coordinate System Conventions:* ANTs internally operates in the LPS (Left-Posterior-Superior) convention; the conversion between RAS and LPS is handled automatically during standard NIfTI file loading within the ANTs framework. An early implementation identified a systematic left-right flip caused by supplying RAS-encoded spatial metadata directly to an interface that expected LPS-encoded values, stressing the importance of using the standard file-loading pathway that performs convention conversion automatically.

G. Protocol Administration and Complementarity

The two protocols assess distinct and non-interchangeable aspects of report quality, and both are applied to each case. The rubric-based protocol produces a weighted composite score, a narrative justification, and a set of safety-critical error flags and is completed only by the five human evaluators (three for the open-source datasets and two for the clinical dataset). The F_1 -based protocol provides Precision, Recall, F_1 , and their per-category breakdowns and is completed only by the LLM judge. The human protocol explains the reasons for each failure

through explicit criterion-level scoring and free-text justification, while the LLM protocol quantifies the extent of errors using finding-level inventories and reproducible aggregate metrics. In the following section, we present both sets of metrics for all three datasets and examine their agreement.

APPENDIX B EXTENDED RESULTS

A. In-House Cohort, LLM-as-Judge: Distribution and Per-Category Performance

In-house institutional dataset — automated LLM-as-judge evaluation ($n = 136$ patients)

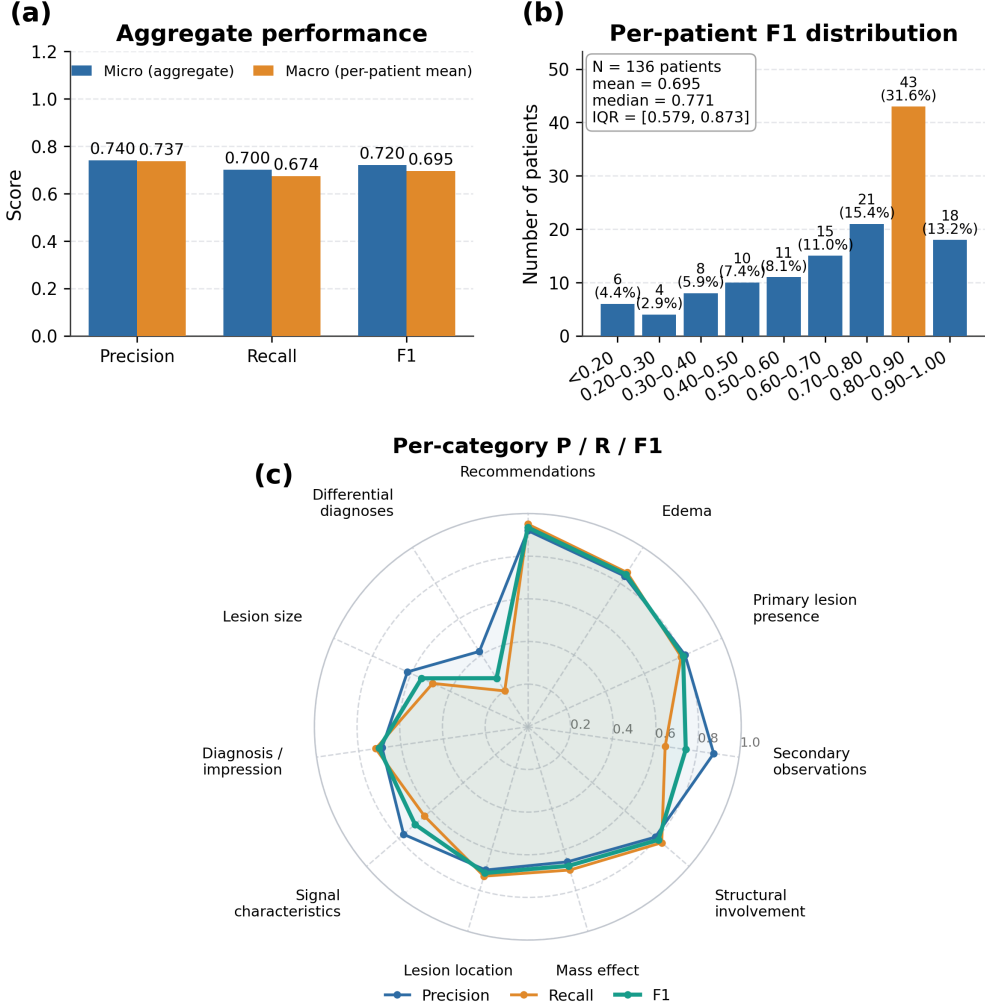


Fig. 5. (a) Aggregate performance of the LLM-as-judge protocol on the in-house institutional dataset ($n = 136$ patients): micro-averaged scores (pooled over TP = 3118, FP = 1096, FN = 1334; blue) against macro-averaged scores (mean of per-patient triples; orange). (b) Distribution of per-patient F1 scores, in bins of width 0.10 with a separate left bin for $F_1 < 0.20$; the peak interval $[0.80, 0.90]$ is highlighted (mean 0.695, median 0.771, IQR $[0.579, 0.873]$). (c) Per-category Precision, Recall and F1 across the eleven finding categories of Section II-G3 of the main paper. This figure also appears as Fig. 3 of the main paper.

1) *Micro- versus macro-averaging*: The micro-macro gap is largest on Recall (0.026) and smallest on Precision (0.003), which indicates that the cases with the most omissions tend to have shorter reference reports, so that those omissions are amplified once every patient contributes equally to the score.

2) *Per-patient distribution*: The per-patient F1 distribution is right-skewed (Fig. 5b). The median patient F1 of 0.771 exceeds the mean of 0.695, and the interquartile range $[0.579, 0.873]$ shows that 75% of all patients lie above $F_1 = 0.58$. The peak interval is $[0.80, 0.90]$, containing 43 of 136 patients (31.6%), with an additional 18 patients

(13.2%) in $[0.90, 1.00]$ and 21 patients (15.4%) in $[0.70, 0.80]$; in aggregate, 82 of 136 patients (60.3%) achieve an F_1 of at least 0.70. The lower tail is correspondingly thin: only 18 patients (13.2%) fall below $F_1 = 0.40$ and just 6 patients (4.4%) below $F_1 = 0.20$. The skew between mean and median, combined with this long but sparse lower tail, implies that a small number of difficult cases lower the mean.

3) *Per-category performance*: The radar plot in Fig. 5c breaks down the aggregate metrics into the 11 finding categories of Section II-G3, and three regimes are visible.

The first regime corresponds to categories on which the pipeline is essentially saturated. Recommendations exceed $F_1 \approx 0.93$, while primary lesion presence, edema, and structural involvement all exceed $F_1 \approx 0.74$ with precision and recall jointly high. These categories are governed by deterministic upstream evidence: the segmentation mask answers presence directly, FLAIR hyperintensity drives the edema flag, and the atlas-based region analysis grounds structural involvement, so the modality-view and view-level agents are largely relaying a well-grounded signal rather than synthesizing new information.

The second regime contains categories with high precision but markedly lower recall. Secondary observations are the clearest example: precision approaches 0.88 while recall falls to roughly 0.65, yielding an asymmetric polygon biased outward on the precision axis. The pipeline rarely fabricates incidental findings, but it routinely omits findings that are present in the radiologist-authored reference report. Typically, these are findings located outside the primary lesion bounding box. Such findings are visible to a human reader who reviews the entire volume, but they fall below the reporting threshold applied by the agents.

The third regime contains categories with depressed precision and recall. Differential diagnoses is the weakest category, with F_1 of approximately 0.27 and recall collapsing to roughly 0.20, indicating the open-ended nature of differential lists in which the agents rarely recover the alternatives enumerated by the reference radiologist. Lesion size is the second-weakest, with F_1 in the neighborhood of 0.55 and recall noticeably lower than precision. This deficit is consistent with the metrological ambiguity of size reporting in clinical practice: reference radiologists variably report a single maximum diameter, three orthogonal diameters, or a volumetric estimate. None of these approaches is uniformly standardized across the dataset, which makes semantic matching of findings more challenging in this category compared to those linked to discrete categorical labels. Mass effect, lesion location, signal features, and the diagnosis-or-impression category occupy an intermediate band at roughly $F_1 \in [0.60, 0.75]$, with no systematic precision-recall asymmetry across the group.

B. In-House Cohort, Human Rubric: Per-Criterion, Per-Evaluator and Inter-Rater Analyses

1) *Per-criterion averages*: Averaged across all 272 evaluations, the six rubric criteria fall into a clear ordering (Fig. 6a). Clinical utility receives the highest mean score at 7.02/10, followed by diagnostic concordance (6.52) and reporting quality (6.51). The three remaining criteria, factual accuracy (5.84), hallucination and fabrication (5.93), and completeness are the lowest scoring.

2) *Interpretation of the per-criterion ordering*: This ordering is clinically informative: management-level reasoning, captured by clinical utility and diagnostic concordance, is judged stronger than the underlying factual substrate of the report. In particular, the safety-critical criteria (factual accuracy, hallucination, diagnostic concordance), which jointly carry 60% of the composite weight, do not collapse uniformly; diagnostic concordance is in fact among the strongest criteria, while factual accuracy and hallucination are the joint weakest. This pattern is consistent with a pipeline that occasionally introduces fine-grained factual deviations yet preserves the overall diagnostic framing of the case.

3) *Per-evaluator decomposition*: The 12-point composite gap between Evaluator A and Evaluator B is not uniformly distributed across criteria (Fig. 6b). The largest per-criterion gap is on reporting quality (+1.84 in favor of Evaluator B, 7.43 vs. 5.59), followed by hallucination and fabrication (+1.77, 6.81 vs. 5.04) and factual accuracy (+1.60, 6.64 vs. 5.04). The smallest gap, in contrast, is on clinical utility (+0.20, 7.12 vs. 6.92): the two raters are in close agreement that the pipeline’s management recommendations are clinically reasonable, even though they disagree on the faithfulness of the underlying narrative. Diagnostic concordance shows the second-smallest gap (+0.79). The implication is that absolute rubric scores must be interpreted relative to the rater, but the qualitative finding holds: clinical utility and diagnostic concordance are the strongest dimensions and factual accuracy and hallucination the weakest. This pattern is consistent across raters.

In-house institutional dataset — human rubric evaluation (272 paired evaluations)

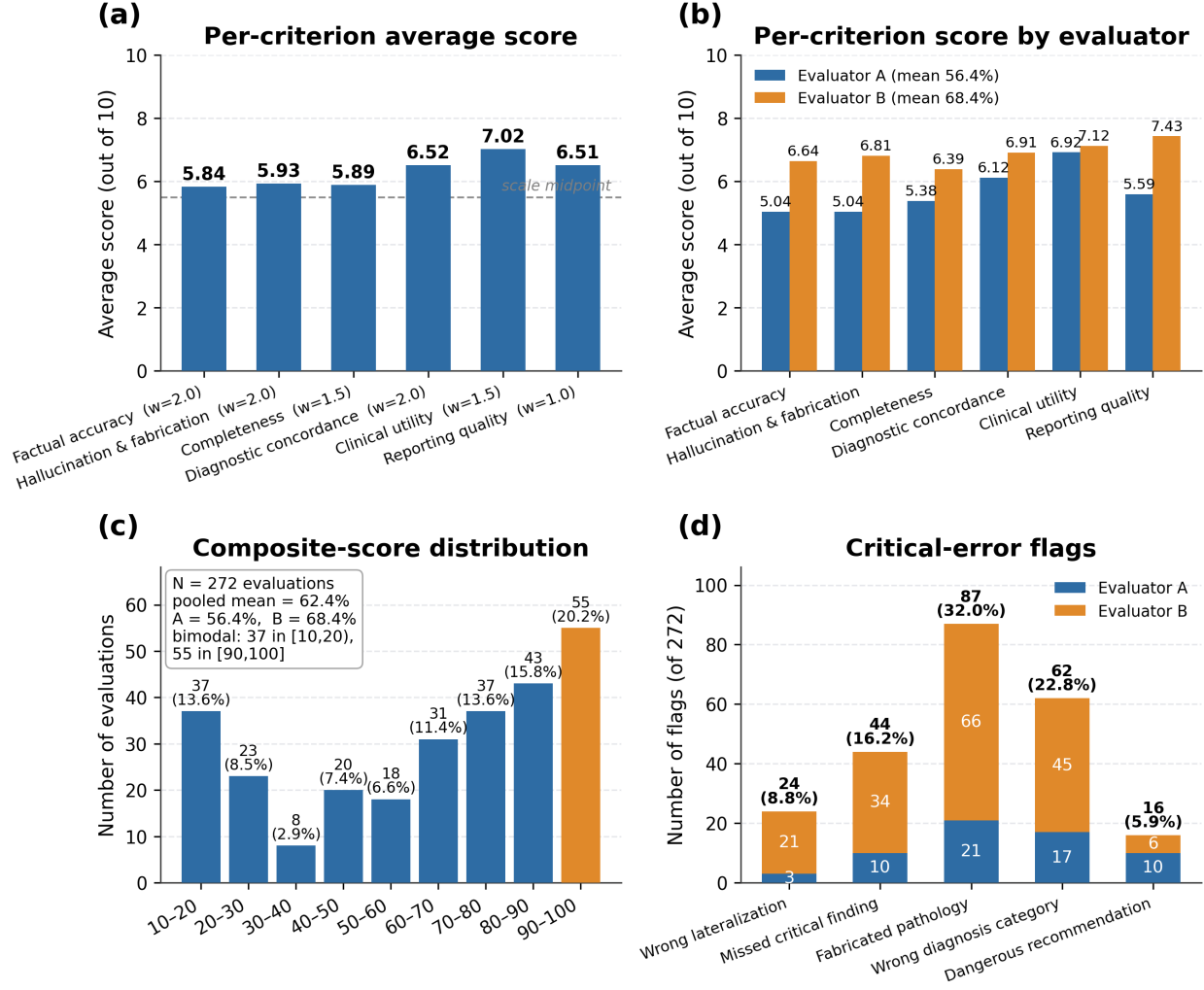


Fig. 6. (a) Per-criterion average rubric scores on the in-house institutional dataset ($n = 136$ patients, 272 paired evaluations), with the criterion weights w_i of (2) annotated beneath each criterion name. (b) The same scores stratified by evaluator. (c) Distribution of the 272 weighted composite scores, in bins of width 10 percentage points; the distribution is bimodal, with 49.6% scoring at least 70% and 25.0% below 40%. (d) Critical-error flag counts by category, stacked by evaluator. This figure also appears as Fig. 4 of the main paper.

4) *Distribution of composite scores*: The distribution of the 272 weighted composite scores is markedly bimodal (Fig. 6c). A large upper mode contains 135 evaluations (49.6%) scoring at least 70%, dominated by the [90, 100] bin with 55 evaluations (20.2%); a smaller but non-trivial lower mode contains 68 evaluations (25.0%) scoring below 40%, with the [10, 20) bin alone accounting for 37 evaluations (13.6%). The intermediate region [40, 70) holds the remaining 69 evaluations (25.4%). This bimodal shape mirrors the right-skewed pattern observed in the LLM-judge per-patient F1 distribution (Section III-A): in both protocols, performance is not driven by uniformly mediocre reports but by a sharp separation between cases where the pipeline performs well and a smaller subset of cases where it fails substantially.

5) *Per-rater breakdown of the critical-error flags*: The most clinically reassuring finding is that dangerous-recommendation flags are the rarest, with only 16 instances (5.9%) across all evaluations. Even when a report is judged factually or diagnostically flawed, the recommended management is rarely considered likely to cause patient harm if followed. The per-rater breakdown shows that Evaluator B raised substantially more flags than Evaluator A on every category *except* dangerous-recommendation, where the ordering reverses (10 for A vs. 6 for B); this is consistent with Evaluator A having a lower threshold for flagging management-level risk specifically, even while assigning systematically lower composite scores overall.

6) *Clinical usefulness rating*: In addition to the rubric, each evaluator gave a binary verdict on whether the AI-generated report would be useful in a clinical workflow. Across the 272 evaluations, 149 (54.8%) were marked useful and 123 (45.2%) not useful. The two raters concur closely on this binary judgment, Evaluator A 53%, Evaluator B 57% — despite their substantially different composite-score levels, which suggests that the usefulness verdict is governed by a coarse pass/fail threshold, rather than by the detailed criteria that determine the rubric composite score.

7) *Inter-rater agreement*: With both raters reviewing all 136 patients, the rubric instrument supports direct inter-rater analysis. The mean per-patient standard deviation of the weighted composite score is 11.45 percentage points (mean range 16.19, maximum range 85.0), indicating that raters typically disagree by roughly a tenth of the scale. However, a small number of cases produced near-maximal disagreement. At the criterion level, the mean per-patient standard deviations on the 1-10 scale are: diagnostic concordance 1.19 (lowest), clinical utility 1.33, completeness 1.35, reporting quality 1.46, factual accuracy 1.54, and hallucination and fabrication 1.56 (highest). The pattern is interpretable: raters agree most on the structured, label-like dimensions of diagnostic concordance and clinical utility and disagree most on the open-ended dimensions of hallucination and factual accuracy, where a single ambiguous sentence can be read as either a benign refinement or a fabrication depending on the reviewer’s prior expectations. The human rubric evaluation of the in-house dataset yields a pooled mean composite score of 62.4%, which is qualitatively consistent with the LLM judge’s aggregate F1 of 0.720. Both protocols indicate that the pipeline is clinically useful in most cases, with a smaller but meaningful subset of reports that fail substantively. The protocols agree on two clinically important findings: clinical utility is the strongest rubric criterion, and dangerous-recommendation is the rarest critical flag. They also agree that the rubric’s weakest criteria, factual accuracy and hallucination, correspond to the LLM-judge categories most susceptible to fabrication-style errors. The protocols differ in which error mode is most prominent. The LLM judge identifies omission as the main residual error, while the human rubric more frequently flags fabricated pathology. This difference is partly due to the protocols’ matching criteria. The LLM-judge protocol accepts AI-generated details that refine but do not contradict the reference as matches. In contrast, human reviewers may flag such content as fabrication if the added entity has clinical implications.

8) *Agreement and divergence between the two protocols*: The protocols agree on two clinically important findings: clinical utility is the strongest rubric criterion, and dangerous-recommendation is the rarest critical flag. The protocols differ in which error mode is most prominent: the LLM judge identifies omission as the main residual error, while the human rubric more frequently flags fabricated pathology. This difference is partly due to the protocols’ matching criteria: the LLM-judge protocol accepts AI-generated details that refine but do not contradict the reference as matches, whereas human reviewers may flag such content as fabrication if the added entity has clinical implications.

C. RHUH-GBM: Per-Category Performance

Decomposing the aggregate scores into the eleven finding categories of Section II-G (Fig. 7b) reveals three regimes, consistent in shape but elevated in level relative to the in-house dataset.

The first regime contains eight of the eleven categories at $F_1 \geq 0.93$. Structural involvement and differential diagnoses both are attained $F_1 = 1.000$ with zero false positives and zero false negatives, indicating that for this homogeneous glioblastoma dataset, the atlas-grounded structural analysis and the agents’ differential listing are in complete semantic agreement with the reference. Mass effect ($F_1 = 0.970$), lesion location (0.968), secondary observations (0.967), primary lesion presence (0.964), edema (0.946), and signal features (0.936) populate the remainder of this saturated regime, each with precision at or near unity and recall in the range 0.90–0.99. As in the in-house dataset, these categories are governed by deterministic upstream evidence, the segmentation mask answers presence and edema directly, the atlas-based region analysis grounds location and structural involvement, and the multi-sequence intensity profile constrains signal features. The per-category structure of the LLM judge mirrors the per-module structure of the pipeline.

The second regime contains diagnosis or impression at $F_1 = 0.795$, with precision 1.000 but recall 0.659. The asymmetry indicates that when the pipeline issues a diagnostic claim it is supported by the reference, but it routinely omits secondary diagnoses, IDH/MGMT-status notes, and WHO grade qualifiers that the reference reports record alongside the primary glioblastoma label.

The third regime contains recommendations ($F_1 = 0.584$) and lesion size ($F_1 = 0.489$). Recommendations are highly precise (0.966) but recover only 41.9% of reference findings: the RHUH reports are recommendation-rich,

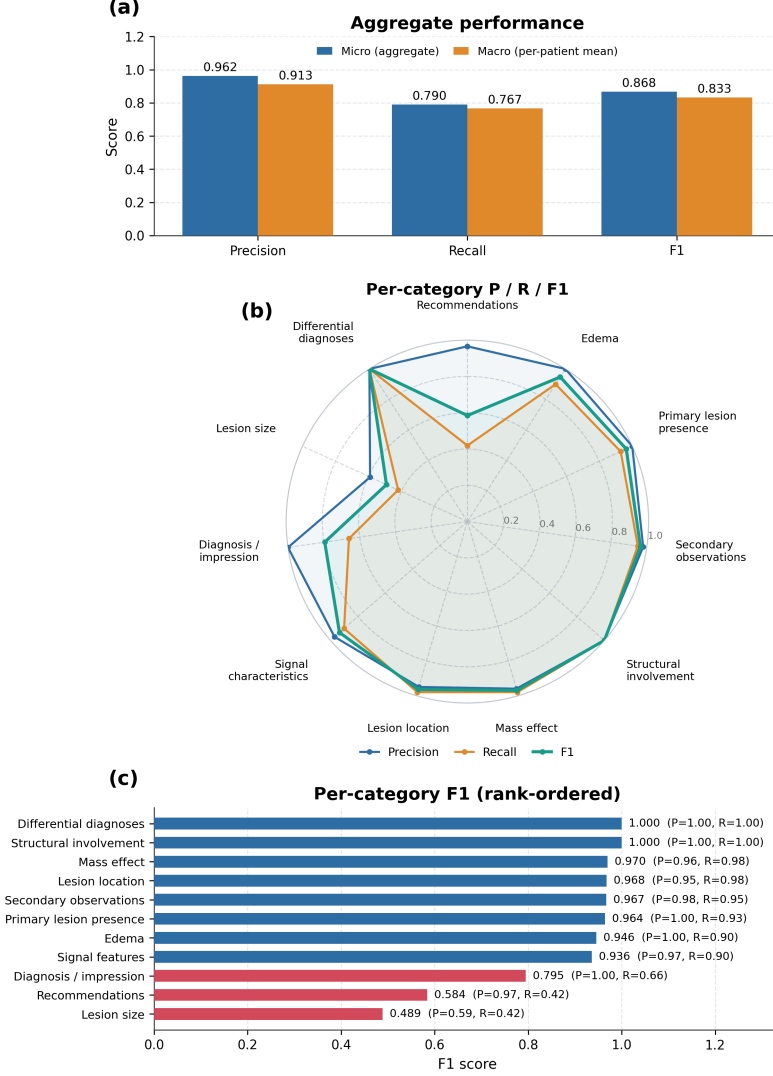
RHUH-GBM dataset — automated LLM-as-judge evaluation ($n = 40$ patients)

Fig. 7. (a) Aggregate performance of the LLM-as-judge protocol on the RHUH-GBM dataset ($n = 40$ patients). Bars compare micro-averaged scores (pooled across all TP, FP, FN counts; blue) with macro-averaged scores (mean of per-patient (Precision, Recall, F1) triples; orange) for Precision, Recall, and F1. Aggregate counts: TP = 1212, FP = 48, FN = 322. (b) Per-category Precision, Recall, and F1 of the LLM-as-judge protocol on the RHUH-GBM dataset, decomposed across the eleven finding categories defined in Section II-G. Each axis ranges from 0 to 1. Eight of eleven categories saturate at $F_1 \geq 0.93$ with near-symmetric Precision and Recall; the residual deficits are concentrated in recommendations, lesion size, and diagnosis or impression. (c) Per-category F1 on the RHUH-GBM dataset, rank-ordered. Bars are annotated with Precision and Recall to make the precision–recall regime of each category visible at a glance; the three weakest categories (lesion size, recommendations, diagnosis or impression) are highlighted in red.

enumerating specific follow-up imaging intervals, multidisciplinary referrals, and surveillance protocols that the agents do not consistently generate. Lesion size is the weakest category on both axes ($P = 0.590$, $R = 0.418$) and reflects the same metrological ambiguity observed in the in-house dataset: the RHUH reference quantifies tumors by exact volumes (e.g., 103.21 cm^3), whereas the pipeline reports linear diameters or qualitative descriptors, and the semantic matcher conservatively classifies cross-format pairs as partial misses rather than as matches.

1) *Cross-dataset comparison with the in-house dataset:* The RHUH-GBM results sit substantially above the corresponding in-house results of Section III-A of the main paper on every aggregate axis: micro F1 rises from 0.720 to 0.868 (+0.148), micro precision rises from 0.740 to 0.962 (+0.222). The largest per-category swing is on differential diagnoses, which collapses to $F_1 \approx 0.27$ on the in-house dataset but saturates at $F_1 = 1.000$ on RHUH. We attribute this gap to two structural differences between the datasets rather than to a difference in pipeline

behavior. First, RHUH is a homogeneous, treatment-naive glioblastoma dataset whose reference reports were curated for this study under a single consistent template (Section II-B of the main paper), whereas the in-house reports are routine free-text clinical dictations. Second, several in-house cases have post-operative or normal-brain references against which the AI report nonetheless describes an active tumor, producing genuine clinical mismatches that the judge correctly classifies as fabrications.

D. UCSF-PDGM: Coverage, Per-Criterion, Per-Evaluator, Distribution, Critical Flags and Inter-Rater Agreement

TABLE II

PER-EVALUATOR COVERAGE OF THE UCSF-PDGM COHORT. OF THE 458 PATIENTS IN THE EVALUATION SET, 445 (97.2%) WERE SCORED, YIELDING 865 REPORT-LEVEL EVALUATIONS ACROSS THE THREE RATERS. COVERAGE IS THE PERCENTAGE OF THE 445 EVALUATED PATIENTS REVIEWED BY EACH RATER.

Evaluator	Patients reviewed	Coverage
Evaluator 1	437	98.2%
Evaluator 2	216	48.5%
Evaluator 3	212	47.6%

1) *Characterisation of the outlier rater:* Evaluator 2 is thus a clear outlier, rather than simply existing at the far end of a normal distribution of strictness: the two agreeing raters agree closely (pairwise mean standard deviation of 7.2 percentage points on jointly reviewed cases), while the gap to Evaluator 2 is both large and uniform. We report per-rater breakdowns throughout.

2) *Per-criterion averages:* Averaged across all 865 evaluations, the six rubric criteria fall within a narrow and uniformly high band (Fig. 8a). Reporting quality receives the highest mean score at 8.56/10, followed by diagnostic concordance and clinical utility (both 8.35); completeness (8.14), hallucination and fabrication (8.08), and factual accuracy (7.96) complete the ordering. As on the in-house dataset, the two safety-critical criteria of factual accuracy and hallucination are the joint weakest, while diagnostic concordance remains among the strongest. This is again consistent with a pipeline that occasionally introduces fine-grained factual deviations yet maintains the overall diagnostic framing of the case, as now observed on a larger and independently reviewed dataset.

3) *Per-evaluator decomposition:* The outlier behavior of Evaluator 2 is uniform across criteria (Fig. 8b). Evaluator 2 scores every criterion in the range 4.5 to 5.2, between roughly three and a half and five points below Evaluator 1 and Evaluator 3, which cluster between 8.4 and 10.0 on all six criteria. The gap is therefore broadly constant across dimensions rather than concentrated in any single criterion, which indicates a global difference in scoring stringency rather than a disagreement about a specific aspect of report quality. Evaluator 1 is highest on every criterion and reaches the ceiling on reporting quality (10.0), and the within-rater ordering of criteria is consistent across all three raters even though their absolute levels differ. As on the in-house dataset, absolute rubric scores must be interpreted relative to the individual rater.

4) *Distribution of composite scores:* The distribution of the 865 composite scores is strongly concentrated at the top of the scale (Fig. 8c). The modal bin [90, 100] alone contains 525 evaluations (60.7%), and only 13 evaluations (1.5%) fall below a composite of 30. Between these extremes the distribution exhibits a secondary shoulder: 181 evaluations (20.9%) lie in the [30, 60) band, almost entirely attributable to the strict Evaluator 2. The shape echoes the separation observed on the in-house dataset, where performance was driven by a sharp split between a large set of strong reports and a smaller set of weak ones, but the upper mode is far more dominant in this dataset, and the lower shoulder is attributable to a single rater.

5) *Critical error flags:* The five binary critical-error flags follow the same rank ordering observed on the in-house dataset (Fig. 8d). Across the 865 evaluations, fabricated pathology is flagged most frequently (282 evaluations, 32.6%), followed by missed critical finding (86, 9.9%), wrong diagnosis category (72, 8.3%), and wrong lateralization (11, 1.3%). Dangerous recommendation is the rarest flag, raised in only 7 evaluations (0.8%); even where a report is judged factually or diagnostically flawed, the recommended management is rarely considered capable of causing patient harm if followed. The flags are heavily concentrated in the outlier rater: Evaluator 2 alone accounts for 207 of the 282 fabricated-pathology flags (73%) and 66 of the 72 wrong-diagnosis-category flags (92%), reinforcing that this rater applies a substantially lower threshold for flagging across the board.

UCSF-PDGM dataset — human rubric evaluation (865 evaluations, 3 raters)

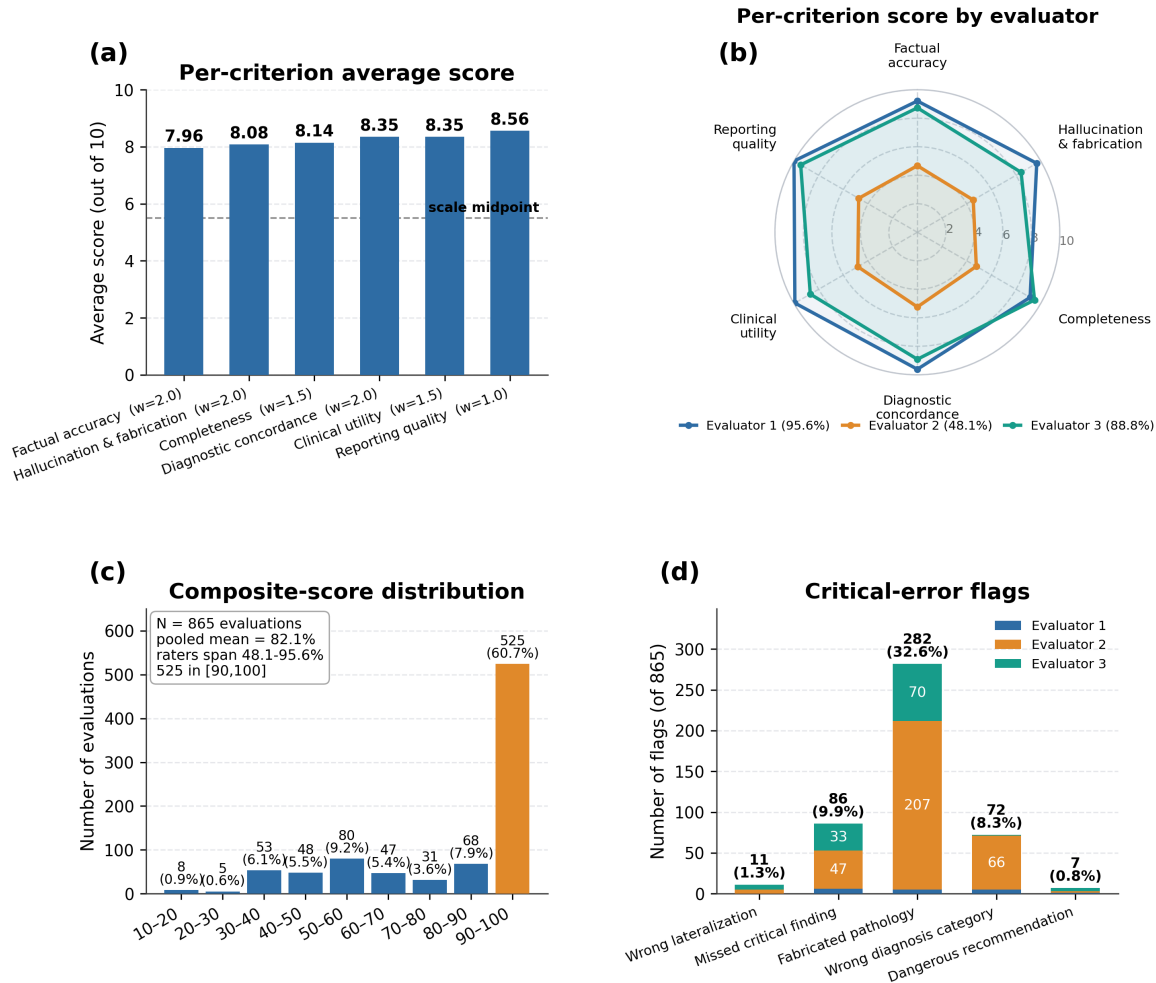


Fig. 8. (a) Per-criterion average rubric scores on the UCSF-PDGM dataset (445 patients, 865 evaluations). Each bar shows the mean score on the 1 to 10 rubric scale; criterion weights w_i used in the composite (Eq. (2)) are annotated beneath each criterion name. (b) Radar plot of the per-criterion scores for each of the three evaluators, with each rater's mean composite in parentheses and every axis on the 1 to 10 scale. Evaluator 2 (orange) traces an inner contour roughly three and a half to five points below the near-saturated polygons of Evaluator 1 (blue) and Evaluator 3 (teal) on every criterion, indicating a uniform difference in stringency rather than a criterion-specific disagreement. (c) Distribution of the 865 weighted composite scores, in bins of width 10 percentage points over the achievable range [10, 100] with the modal bin [90, 100] highlighted; the distribution is concentrated at the top of the scale, with a smaller secondary shoulder in [30, 60] almost entirely contributed by Evaluator 2. (d) Critical-error flag counts by category, stacked by evaluator; percentages above each bar are computed against the 865 evaluations. The dangerous-recommendation category, the most safety-relevant flag, is the rarest overall and heavily concentrated in the outlier Evaluator 2.

6) *Clinical usefulness by evaluator*: The verdict again separates the outlier from the rest: Evaluator 1 and Evaluator 3 marked 100% and 83% of their cases useful respectively, whereas Evaluator 2 marked only 11% useful.

7) *Inter-rater agreement*: Restricting attention to the 257 patients reviewed by more than one of the three raters, the mean per-patient standard deviation of the weighted composite score is 22.7 percentage points (mean range 38.0, maximum range 79.5), roughly double the 11.45-point dispersion observed on the in-house dataset. This inflated dispersion is itself largely an artifact of the outlier: agreement between the two concordant raters alone is far tighter (pairwise mean standard deviation of 7.2 points), so most of the apparent disagreement on this dataset reflects the gap between Evaluator 2 and the other two rather than diffuse noise. At the criterion level, the mean per-patient standard deviations on the 1 to 10 scale are lowest for diagnostic concordance (2.20) and factual accuracy (2.26) and highest for clinical utility (2.44) and hallucination (2.43); as on the in-house dataset, diagnostic concordance shows the tightest agreement and hallucination is among the widest. Taken together, the UCSF-PDGM results confirm,

on the largest and most heavily reviewed dataset, the two findings that were consistent across the in-house and RHUH evaluations: clinical management content is judged at least as reliable as the underlying factual detail, and dangerous recommendations remain the rarest safety-critical failure.

E. Extended Conclusion

By assigning specialized agents to individual sequence–plane combinations, reconciling their observations across orthogonal views and grounding the resulting narrative in segmentation, atlas localization and quantitative image measurements, the system makes intermediate evidence accessible for inspection.

The results identify both the promise and the present limits of this approach. BrainAgent reproduced lesion presence, location, oedema, structural involvement and clinically appropriate management content with comparatively high consistency, and generated reports considered useful in a substantial proportion of cases. However, performance declined for lesion size, incidental observations and open-ended differential diagnosis, and a subset of cases showed major factual or diagnostic failures. The contrast between strong results on homogeneous preoperative glioblastoma data and more moderate performance in the real-world cohort highlights the sensitivity of report-generation systems to disease heterogeneity, postoperative appearances, acquisition variability and differences in reference-report conventions. Moreover, disagreement among evaluators, particularly in judging hallucination and factual accuracy, shows that evaluation standards themselves remain an important source of uncertainty.

F. Author Contributions

A. T. Elboardy: Data curation, Formal analysis, Software, Visualization, Writing – original draft; **Z. Elshaer:** Data curation, Software, Visualization; **S. Ahmed:** Data curation, Validation; **I. Shehaby:** Data curation, Validation; **A. S. E. Raslan:** Data curation, Validation; **K. K. Elfatairy:** Investigation, Validation; **N. A. Tawfik:** Investigation, Validation; **M. Al-Shatouri:** Investigation, Validation; **G. Khoriba:** Conceptualization, Investigation, Methodology, Supervision, Writing – review & editing; **M. A. Mabrok:** Conceptualization, Investigation, Methodology, Supervision, Writing – review & editing; **E. A. Rashed:** Conceptualization, Investigation, Funding acquisition, Methodology, Project administration, Supervision, Writing – review & editing.

APPENDIX C REPRODUCIBILITY RECORD

This appendix provides a reproducible record of (a) every prompt supplied to the report-generation agents, the consolidation orchestrators, and the large-language-model (LLM) judge, and (b) the models and hyperparameters used for report generation, automated evaluation, and image analysis. Brace tokens such as {patient_id} are runtime placeholders interpolated when the prompt is rendered. Implementation-level source code is released separately with the accompanying repository.

A. Multi-Agent Architecture

Report generation uses a three-tier hierarchical multi-agent design with a deterministic axial-reference consolidation rule.

- **Tier 1: per-modality/per-view writer agents.** One agent per (modality $\times$ view) combination, with modalities {T1, T1c, T2, FLAIR} and views {Axial, Coronal, Sagittal}, giving twelve agents. Each agent produces one initial report and then performs iterative vision-refinement passes over successive image batches.
- **Tier 2: view-level consolidation agents ($\times 3$).** One per anatomical plane. Each fuses the four modality reports acquired in its plane into a single coherent view-level report.
- **Tier 3: master orchestrator ($\times 1$).** Fuses the axial, coronal, and sagittal view reports into the final report, treating the axial report as the primary reference and resolving contradictions in its favour unless another plane provides unambiguous complementary evidence.

An analysis-mode switch selects the prompt set per patient: when the tumor volume is below 1 cm³ (or no mask is produced), the agents use the *General-Brain* prompt set; otherwise they use the *Tumor-Analysis* prompt set. The visual input to each Tier-1 agent is a sequence of batch grid images, each tiling six slices in a 3 $\times$ 2 layout, which balances anatomical coverage against the input-resolution limits of the vision model. Reasoning (“thinking”) mode

TABLE III
END-TO-END PER-PATIENT PROCESSING FLOW

Stage	Description
Input handling	File grouping by patient; DICOM-to-NIfTI conversion where required
Preprocessing	Reorientation to RAS+, N4 bias-field correction, 1 mm isotropic resampling
Skull stripping	Optional removal of non-brain tissue
Tumor segmentation	ConvNextLocator detection/localization with MedSAM2 refinement
MNI-152 registration	SyN registration to the ICBM152 template
Region analysis	Atlas-based labelling (Harvard-Oxford and Pauli)
Tumor metrics	Volume, area, centroid, and hemisphere determination
Visual assembly	2D slice extraction and batch-grid image construction
Report generation	Three-tier multi-agent generation (this appendix)
Output	Final report and 3D surface meshes

is disabled for the twelve Tier-1 image-analysis agents and enabled for the Tier-2 and Tier-3 consolidation agents, where cross-modal and cross-planar synthesis benefits from it. Table III summarizes the end-to-end per-patient processing flow.

B. Models

All twelve writer agents, the three view-level consolidation agents, and the master orchestrator share a single vision-language backbone, Qwen3.5-35B-A3B (a mixture-of-experts model). The finding-level F1 analysis is run with a single LLM judge, gpt-5.4; the six-criterion rubric is completed by the human evaluators (see the Evaluation Protocol section of the main text), not by the LLM. The remaining entries in Table IV are the supporting image-analysis models. The agentic architecture is backbone-agnostic, so the vision-language model can be replaced without changing the pipeline.

C. Generation and Sampling Settings

Table V reports the decoding settings for each agent tier. Nucleus sampling uses $\text{top-}p = 0.95$; no $\text{top-}k$, repetition, presence, or frequency penalties are applied, and no fixed decoding seed is set, so reproducibility relies on the low sampling temperatures together with the fixed registration seed. The LLM judge runs at temperature 0.1.

TABLE IV
MODELS USED FOR GENERATION, EVALUATION, AND IMAGE ANALYSIS.

Role	Model	Notes
Generation backbone (all agents)	Qwen3.5-35B-A3B	MoE vision-language model
Finding-level F1 judge	gpt-5.4	Atomic-finding TP/FP/FN classification
Tumor detection/localization	ConvNextLocator	ConvNeXt-Tiny backbone with dual head
Mask refinement	MedSAM2 (SAM2.1 Hiera-Tiny)	Box-prompted segmentation
Registration	ANTs SyN	Nonlinear MNI-152 registration
Skull stripping	HD-BET / ANTs / SynthStrip	Selectable; HD-BET default
Registration target	MNI ICBM152 2009a T1	Fixed template
Cortical atlas	Harvard-Oxford	≈ 48 cortical regions
Subcortical atlas	Pauli 2017	16 subcortical nuclei

D. Report-Generation Agent Prompts

Two short system messages frame the Tier-1 writer-agent calls.

System Message A.1 Initial-report system message

You are an expert radiologist with extensive experience in multi-modal medical imaging
 → analysis, anatomical interpretation, and automated image analysis integration.

TABLE V
PER-TIER GENERATION SETTINGS. "REASONING" DENOTES THE CHAIN-OF-THOUGHT MODE.

Agent tier	Temperature	Max tokens	Reasoning
Tier 1 initial report	0.3	2500	off
Tier 1 vision refinement	0.2	2500	off
Tier 2 view consolidation	0.2	20000	on
Tier 3 master orchestrator	0.2	20000	on

IMAGE ORIENTATION: All images use RADIOLOGICAL convention - the patient's RIGHT side appears
 ↳ on the viewer's LEFT, and the patient's LEFT side appears on the viewer's RIGHT. 'R'
 ↳ and 'L' markers on images confirm this convention. When correlating visual
 ↳ observations with the provided region analysis data, trust the automated
 ↳ lateralization labels ('Left'/'Right' prefixes on region names) as these were
 ↳ computed from MNI-152 atlas coordinates.

System Message A.2 Refinement system message

You are an expert radiologist specializing in brain tumor analysis with extensive experience
 ↳ in multimodal neuroimaging interpretation.

Tumor-Analysis prompt set:

Prompt A.1 Tumor-Analysis: initial report

You are an expert radiologist generating a comprehensive medical imaging analysis report.
 You have been provided with:

1. Quantitative analysis metadata from a VALIDATED AUTOMATED PIPELINE
2. Automated modality detection results
3. A batch image showing multiple slices of the scan WITH TUMOR SEGMENTATION OVERLAYS

PATIENT ID: {patient_id}

TUMOR SEGMENTATION OVERLAY LEGEND (APPLIED TO ALL IMAGES):

- Light Green overlay: Necrotic/Non-Enhancing Tumor Core (NCR/NET) - Class 1
- Light Yellow overlay: Peritumoral Edema (ED) - Class 2
- Light Blue overlay: GD-Enhancing Tumor (ET) - Class 3

These semi-transparent colored overlays have been applied to help you identify and localize the different tumor components on the MRI slices. The underlying grayscale MRI signal is still visible through the overlays. Use these overlays to correlate the tumor location with the quantitative metrics provided below (Note that the quantitative metrics are derived
 ↳ from the validated automated pipeline).

AUTOMATED ANALYSIS RESULTS:

{organ_info}
 {modality_info}

QUANTITATIVE ANALYSIS SUMMARY (from validated automated pipeline - TRUST THESE VALUES):

- Tumor Metrics: {tumor_metrics_json}
- Region Analysis: {region_analysis_json}
- Common Locations: {common_locations}

IMPORTANT: The tumor metrics and region analysis above were computed through a validated
 ↳ automated

image processing pipeline using MNI-152 registration and Harvard-Oxford/Pauli atlas mapping. These quantitative measurements are accurate and should be trusted. Your role is to

↳ interpret

these findings in clinical context and correlate them with the visual observations from the
 ↳ image.

REPORTING GUIDELINES FOR QUANTITATIVE DATA:

- Report the tumor VOLUME (in cm3) as a key measurement
- List the affected brain regions with their involvement percentages
- Do NOT include centroid coordinates (X, Y, Z) in the report - these are internal pipeline
 ↳ values
- Focus on clinical interpretation: which regions are affected and what that means
 ↳ clinically

IMAGING CONTEXT:

{context_info_json}

PATIENT & STUDY INFORMATION (from DICOM metadata):

{metadata_summary_json}

NOTE: The metadata_summary above contains:

- patient_info: patient demographics (age, sex, study date)
- scanner_info: MRI scanner details (manufacturer, model, field strength)
- tumor_metrics: automated analysis results
- region_analysis: brain region involvement data

VISUAL ANALYSIS INSTRUCTIONS:

- Carefully analyze the provided batch image showing multiple slices WITH COLORED OVERLAYS
- Use the overlay colors to identify different tumor components (necrotic core=green, edema=yellow, enhancing=blue)
 ↳ yellow, enhancing=blue)
- Correlate visual observations with the quantitative metrics provided
- Note any additional findings visible in the images not captured by automated analysis
- Verify that visual observations are consistent with the reported tumor location and extent
- Describe the spatial relationship between the different tumor components visible through
 ↳ the colored overlays

Please generate a comprehensive initial radiology report including:

1. CLINICAL HISTORY & INDICATION

- Include detected organ system and imaging approach

2. TECHNIQUE & METHODOLOGY

- Note automated analysis methods used

3. FINDINGS:

- Organ-specific observations based on detected anatomy and which areas of the brain were
 ↳ affected by the tumor
- Tumor Characteristics (size, location, morphology) - correlate with provided metrics
- Tumor Class Analysis (ET, ED, NCR/NET distribution)
- Spatial Distribution Analysis
- Visual observations from the batch image
- Modality-specific findings

4. IMPRESSION

- Integrate automated detection results with quantitative analysis and visual findings
- Consider organ-specific differential diagnoses

5. RECOMMENDATIONS

- Organ and modality-appropriate follow-up suggestions

IMPORTANT CONSIDERATIONS:

- Use the detected organ type ({primary_organ}) to guide anatomical interpretation
- Incorporate the identified modalities ({detected_modalities_list}) in technical
 ↳ descriptions
- Maintain professional medical language and standard radiology terminology
- The region analysis is comprehensive - address the major affected regions of the brain and
 ↳ classify the tumor based on this information

- DO NOT contradict the quantitative measurements unless you observe a clear discrepancy in
 ↪ the images
- Briefly note the status of the following structures (you should comment about them): bone,
 ↪ brain stem, cerebellum, ventricles, pituitary gland, and vessels. If these
 ↪ structures appear normal, a single sentence noting their normal appearance is
 ↪ sufficient. Do NOT fabricate detailed descriptions of structures you cannot clearly
 ↪ evaluate from the provided images.

Prompt A.2 Tumor-Analysis: iterative refinement

You are an expert radiologist performing iterative improvement of a medical imaging analysis
 ↪ report.

PATIENT CONTEXT:

- Primary organ: {detected_organ}
- Detected modalities: {detected_modalities_list}

TUMOR SEGMENTATION OVERLAY LEGEND (APPLIED TO ALL IMAGES):

- Light Green overlay: Necrotic/Non-Enhancing Tumor Core (NCR/NET) - Class 1
- Light Yellow overlay: Peritumoral Edema (ED) - Class 2
- Light Blue overlay: GD-Enhancing Tumor (ET) - Class 3

The MRI slices have semi-transparent colored overlays showing tumor segmentation. Use these to identify and localize different tumor components while observing the underlying MRI
 ↪ signal.

CURRENT REPORT STATUS:

{current_report}

IMPROVEMENT STAGE: {stage_number}

CURRENT BATCH: {image_name}

CRITICAL CONTEXT ABOUT QUANTITATIVE DATA:

The tumor metrics (volume) and region analysis (affected brain regions) were already
 ↪ provided in the
 initial report generation. These values were computed through a VALIDATED AUTOMATED IMAGE
 ↪ PROCESSING
 PIPELINE using MNI-152 registration and Harvard-Oxford/Pauli atlas mapping.

DO NOT modify, contradict, or re-estimate these quantitative values. They are accurate
 ↪ measurements

from the automated pipeline. Your task is to REFINE THE VISUAL DESCRIPTIONS AND CLINICAL
 ↪ INTERPRETATION

based on the new batch of images, while keeping the quantitative data consistent.

REMINDER: Report tumor volume and affected regions with percentages. Do NOT add centroid
 ↪ coordinates to the report.

INSTRUCTIONS:

1. Carefully analyze the provided medical imaging batch WITH COLORED OVERLAYS
2. Use the overlay colors to identify tumor components (green=necrotic, yellow=edema, blue=
 ↪ enhancing)
3. Compare visual findings with the current report
4. Leverage the known organ type ({detected_organ}) and modalities for accurate
 ↪ interpretation
5. Improve the report by:
 - Adding NEW specific visual observations from this batch
 - Describing the spatial distribution of different tumor components (green, yellow, blue
 ↪ regions)
 - Enhancing descriptions based on organ-specific anatomy visible in this batch
 - Refining qualitative descriptions (e.g., signal characteristics, enhancement patterns)
 - Adding modality-specific clinical correlations
 - Incorporating organ-appropriate differential considerations

6. MAINTAIN the overall structure and ALL quantitative measurements from the initial report
7. Remove "INITIAL REPORT - PENDING VISUAL CONFIRMATION" if still present
8. Add specific visual observations from this batch
9. Integrate findings coherently with previous observations

FOCUS AREAS FOR {detected_organ} IMAGING:

- Organ-specific anatomical landmarks and normal variants
- Pathology patterns typical for this organ system
- Modality-specific signal characteristics and enhancement patterns
- Spatial relationships relevant to this anatomical region
- Clinical significance in the context of this organ system
- Briefly note the status of the following structures (you should comment about them): bone,
 - ↳ brain stem, cerebellum, ventricles, pituitary gland, and vessels. If these
 - ↳ structures appear normal, a single sentence noting their normal appearance is
 - ↳ sufficient. Do NOT fabricate detailed descriptions of structures you cannot clearly
 - ↳ evaluate from the provided images.

Provide the COMPLETE IMPROVED REPORT (not just changes). Preserve all quantitative
 ↳ measurements exactly as stated in the current report.

General-Brain prompt set: Used when the tumor volume is below 1 cm³ or no tumor is detected.

Prompt A.3 General-Brain: initial report

You are an expert neuroradiologist generating a comprehensive brain MRI analysis report.

IMPORTANT CLINICAL CONTEXT:

The automated tumor detection pipeline did NOT identify a significant tumor mass in this
 ↳ scan.

This could mean:

1. The brain appears normal with no detectable lesions
2. Any abnormalities are subtle or non-specific
3. The findings may represent normal variants, artifacts, or incidental findings

YOUR TASK: Perform a THOROUGH general brain analysis looking for ANY abnormalities, not just
 ↳ tumors.

PATIENT ID: {patient_id}

PATIENT & STUDY INFORMATION (from DICOM metadata):
 {metadata_summary_json}

NOTE: The metadata_summary above contains:

- patient_info: patient demographics (age, sex, study date)
- scanner_info: MRI scanner details (manufacturer, model, field strength)
- tumor_metrics: automated detection results (volume was below threshold)

IMAGING PARAMETERS:
 {context_info_json}

VISUAL ANALYSIS INSTRUCTIONS:

You are being shown a comprehensive set of brain MRI slices from the {modality_name}
 ↳ sequence.

Since no significant tumor was detected by automated analysis, carefully examine:

1. BRAIN PARENCHYMA:

- Gray matter / white matter differentiation
- Any focal or diffuse signal abnormalities
- Presence of encephalomalacia, gliosis, or atrophy
- Symmetry between hemispheres

2. VENTRICULAR SYSTEM:

- Size and configuration of lateral ventricles
- Third and fourth ventricles
- Any hydrocephalus or asymmetry
- Periventricular signal changes

3. EXTRA-AXIAL SPACES:

- Sulci and cisterns (prominent? effaced?)
- Subdural or epidural collections
- Meningeal enhancement (if contrast used)

4. VASCULAR STRUCTURES:

- Major arteries and venous sinuses
- Any vascular malformations or aneurysms
- Flow voids

5. SKULL AND CALVARIUM:

- Bone marrow signal
- Calvarial thickness and integrity
- Skull base abnormalities

6. POSTERIOR FOSSA:

- Cerebellum (vermis and hemispheres)
- Brain stem (midbrain, pons, medulla)
- Cerebellar tonsils position

7. SELLA AND PARASELLAR REGION:

- Pituitary gland size and signal
- Optic chiasm and nerves
- Cavernous sinuses

8. ORBITS AND PARANASAL SINUSES:

- Globe and optic nerve appearance
- Sinus aeration and mucosal thickening

9. INCIDENTAL FINDINGS:

- White matter hyperintensities
- Choroid plexus cysts
- Pineal cysts
- Arachnoid cysts
- Developmental venous anomalies

Please generate a comprehensive radiology report including:

1. CLINICAL HISTORY & INDICATION

- Note the clinical question and any relevant patient history

2. TECHNIQUE

- Describe the MRI sequences analyzed ({detected_modalities_list})
- Note any technical limitations

3. FINDINGS:

- Systematic evaluation of all brain structures
- Document normal findings as well as any abnormalities
- Be thorough - even if the brain appears grossly normal, document this systematically

4. IMPRESSION

- Summary of significant findings (or confirmation of normal study)
- Any recommendations for additional imaging or clinical correlation

5. RECOMMENDATIONS

- Follow-up if needed
- Clinical correlation suggestions

CRITICAL REMINDER:

- The automated analysis did NOT detect a significant tumor

- Your role is to provide a comprehensive brain evaluation
- Document both normal and abnormal findings
- If the brain truly appears normal, state this confidently
- Look for subtle findings that automated detection might miss

Prompt A.4 General-Brain: iterative refinement

You are an expert neuroradiologist performing iterative improvement of a general brain MRI
 ↳ analysis report.

CONTEXT:

- No significant tumor was detected by automated analysis
- You are analyzing {modality_name} images for any brain abnormalities
- This is improvement stage {stage_number}
- Current batch: {image_name}

CURRENT REPORT STATUS:

{current_report}

MODALITY BEING ANALYZED: {detected_modalities_list}

INSTRUCTIONS:

- Carefully analyze the provided brain MRI slices
- Compare findings with the current report
- Look for ANY abnormalities, not just tumors:
 - Signal changes in white matter
 - Atrophy patterns
 - Vascular abnormalities
 - Extra-axial collections
 - Ventricular abnormalities
 - Skull/bone findings
 - Incidental findings
- Improve the report by:
 - Adding NEW specific observations from this batch
 - Refining descriptions of normal or abnormal structures
 - Ensuring comprehensive coverage of all visible anatomy
 - Adding modality-specific findings (e.g., FLAIR for white matter, T1c for enhancement)
- MAINTAIN the overall structure and completeness of the report
- Document NORMAL findings - stating "normal" is clinically important

FOCUS AREAS FOR THIS BATCH:

- Anatomical structures visible in these slices
- Any subtle signal abnormalities
- Comparison with expected normal appearance
- Documentation of normal structures (equally important as abnormalities)

Provide the COMPLETE IMPROVED REPORT (not just changes).

E. Orchestrator Prompts

Tier-1 structured mini-report.: The Tier-1 agents emit a structured per-modality report validated against a schema (embedded in the prompt as {escaped_schema_json}).

System Message A.3 Tier-1 structured mini-report (system)

You are a board-certified radiologist writing a meticulous, objective, and consistent mini-
 ↳ report for a patient with a brain tumor. ONLY for the assigned modality. Use precise
 ↳ medical terminology. Analyze both the text and the image (if provided). If data is
 ↳ insufficient, explicitly state limitations. Prioritize safety-critical findings.
 ↳ Return a structured JSON that validates against the provided schema.

Prompt A.5 Tier-1 structured mini-report (user)

Modality: {modality}
 Clinical history/indication: {patient_context}
 Clinical question: {clinical_question}
 Comparison: {comparison}
 Modality-specific input (raw text):
 {modality_input}

Instructions:

- Analyze the provided image in conjunction with the text.
- Technique: briefly describe acquisition relevant to this modality.
- Findings: objective bullet points based on text and image; do not include interpretation
 ↳ here.
- Impression: numbered, prioritized, clinically relevant interpretations for THIS modality
 ↳ only.
- Recommendations: follow-up imaging/clinical steps when appropriate; be concise.
- Limitations: note artifacts, motion, limited coverage, contrast timing issues, etc.
- Critical findings: list urgent findings explicitly, if any.

Ensure factual consistency with provided input and history.

Output requirements:

- Return ONLY a valid JSON object conforming to this JSON Schema:
 {escaped_schema_json}
- Do not include any explanatory text or markdown code fences.

Tier-2 view-level consolidation:

System Message A.4 View-level consolidation (system)

You are a senior neuroradiologist writing a consolidated report for a specific anatomical
 ↳ view ({view_name}) of a brain MRI study.

You are provided with reports from multiple MRI sequences (T1, T1c, T2, FLAIR) all acquired
 ↳ in the {view_name} plane.

Your task is to synthesize these modality-specific reports into a single, coherent view-
 ↳ level report.

IMPORTANT CONTEXT:

- Tumor volume and affected brain regions were computed through a VALIDATED AUTOMATED IMAGE
 ↳ PROCESSING PIPELINE using MNI-152 registration and Harvard-Oxford/Pauli atlas mapping.
- These measurements are ACCURATE and should be preserved exactly as reported.
- Your role is to integrate findings across modalities, not to re-estimate measurements.
- Do NOT include centroid coordinates (X, Y, Z) in the report.

SYNTHESIS GUIDELINES:

1. Integrate complementary information from each modality:
 - T1: Anatomical detail, gray/white matter differentiation
 - T1c (contrast-enhanced): Enhancement patterns, blood-brain barrier disruption, tumor
 ↳ vascularity
 - T2: Edema, fluid content, cystic components
 - FLAIR: Periventricular lesions, subtle edema, infiltrative margins
2. Resolve any apparent inconsistencies by considering modality-specific characteristics
3. Highlight findings that are consistently observed across multiple modalities
4. Note any modality-specific findings that provide unique diagnostic information
5. Do NOT mention segmentation overlays, colored overlays (green, yellow, blue), or any
 ↳ visualization artifacts - these are internal pipeline tools, not clinical findings. Use standard
 ↳ clinical terms (necrotic core, enhancing component, peritumoral edema) instead.

Write in professional medical language. Create a comprehensive view-specific report.

Prompt A.6 View-level consolidation (user)

Patient context: {patient_context}
 Anatomical View: {view_name}
 Clinical question: {clinical_question}
 Comparison: {comparison}

Modality reports for {view_name} view (from T1, T1c, T2, FLAIR sequences):
 {modality_reports_json}

Based on the above modality reports for the {view_name} view, write a consolidated report
 ↳ with:

1. REPORT FINDINGS: Single comprehensive paragraph describing all findings observed in this
 ↳ anatomical plane
2. REPORT IMPRESSION: Single paragraph with clinical interpretation specific to observations
 ↳ in this view

Output requirements:

- Return ONLY a valid JSON object conforming to this JSON Schema:
 {json_schema}
- Do not include any explanatory text or markdown code fences.

Tier-3 master orchestrator: The axial report is treated as the reference.

System Message A.5 Master orchestrator (system)

You are the chief neuroradiologist providing the FINAL authoritative interpretation of a
 ↳ brain MRI study.

You are provided with three consolidated view-level reports:

1. AXIAL VIEW REPORT: This is the GOLD STANDARD and PRIMARY reference
2. CORONAL VIEW REPORT: Secondary/supplementary view
3. SAGITTAL VIEW REPORT: Secondary/supplementary view

YOUR ROLE: Synthesize the view reports into ONE UNIFIED clinical narrative. You are NOT
 ↳ comparing
 views or listing findings per-sequence. Write as if you personally reviewed the entire study
 ↳ and
 are dictating a single, cohesive report.

CRITICAL RULES:

1. ****AXIAL IS THE GOLD STANDARD****:
 - The Axial view report is the PRIMARY and most reliable source of information
 - All quantitative measurements from the Axial report should be preserved EXACTLY as
 ↳ stated
 - If Coronal or Sagittal reports CONTRADICT the Axial report -> TRUST THE AXIAL REPORT
 - If Coronal or Sagittal provide COMPLEMENTARY details consistent with Axial -> Include
 ↳ them
2. ****QUANTITATIVE DATA****:
 - Report tumor VOLUME (cm3) this is the only numerical measurement to include
 - List only the TOP 5 (or fewer) MOST affected brain regions by name - no percentages, no
 ↳ coordinates
 - Do NOT include centroid coordinates (X, Y, Z)
 - Do NOT include region involvement percentages
3. ****DO NOT MENTION****:
 - Segmentation overlays, colored overlays, green/yellow/blue regions, or any
 ↳ visualization artifacts
 (these are internal pipeline tools for the AI, NOT clinical findings)
 - Individual MRI sequences by name (T1, T2, FLAIR, T1c) as separate analysis sections -
 ↳ instead,

```

    describe signal characteristics naturally (e.g., "hyperintense on T2/FLAIR", "enhancing
    ↳ on post-contrast")
- Which view (axial, coronal, sagittal) a finding came from - present findings as unified
    ↳ observations
- Raw data like slice indices, voxel counts, or class labels (Class 1, Class 2, etc.)

4. **REPORT STYLE**:
- Write in professional medical language as if dictating the final official radiology
    ↳ report
- Use complete sentences in paragraph form (no bullet points or numbered lists)
- The report should read as a unified interpretation by a single radiologist, not as a
    ↳ collation of sub-reports
- Describe tumor components using standard clinical terms (necrotic core, enhancing
    ↳ component,
    peritumoral edema) NOT overlay colors or segmentation class names

```

Prompt A.7 Master orchestrator (user)

```

Patient context: {patient_context}
Clinical question: {clinical_question}
Comparison: {comparison}

=== AXIAL VIEW REPORT (GOLD STANDARD - PRIMARY REFERENCE) ===
{axial_report_json}

=== CORONAL VIEW REPORT (SUPPLEMENTARY) ===
{coronal_report_json}

=== SAGITTAL VIEW REPORT (SUPPLEMENTARY) ===
{sagittal_report_json}

INSTRUCTIONS:
1. Use the AXIAL report as your primary source of truth
2. Check Coronal and Sagittal reports for ADDITIONAL information that complements the Axial
    ↳ findings
3. If Coronal/Sagittal contradict Axial -> Use Axial findings only
4. Preserve tumor volume exactly as stated in the Axial report
5. List only the top 7 (or fewer) most affected brain regions by name - NO percentages
6. Do NOT mention segmentation overlays, colored regions, or visualization tools
7. Write a UNIFIED narrative - do not attribute findings to specific views or sequences
    ↳ separately

Write your FINAL radiology report with:
1. REPORT FINDINGS: Single comprehensive paragraph describing all findings (Axial as base +
    ↳ supplementary info from other views)
2. REPORT IMPRESSION: Single paragraph with clinical interpretation and recommendations

Output requirements:
- Return ONLY a valid JSON object conforming to this JSON Schema:
{json_schema}
- Do not include any explanatory text or markdown code fences.

```

F. LLM-as-a-Judge: Finding-Level $F1$

The finding-level judge decomposes both the reference and AI reports into atomic findings across eleven categories (primary lesion presence, lesion location, lesion size, signal characteristics, mass effect, edema, structural involvement, secondary observations, diagnosis/impression, differential diagnoses, and recommendations), classifies each finding as true positive, false positive, false negative, or true negative by semantic matching, and reports $\text{Precision} = \text{TP}/(\text{TP} + \text{FP})$, $\text{Recall} = \text{TP}/(\text{TP} + \text{FN})$, and $F_1 = 2\text{Precision} \cdot \text{Recall}/(\text{Precision} + \text{Recall})$. Micro metrics pool counts across patients before applying these formulas; macro metrics average per-patient values; per-category metrics are computed likewise. True negatives are recorded but excluded from the metrics.

Judge Prompt A.1 Finding-level F1 judge: system prompt

You are a senior neuroradiologist serving as an impartial judge for an automated evaluation of AI-generated brain MRI radiology reports. Your task is to convert each report pair into a structured finding-level inventory and classify every finding as True Positive (TP), False Positive (FP), or False Negative (FN) according to the protocol below.

=== ATOMIC FINDING EXTRACTION ===

Decompose BOTH the reference report and the AI report into discrete atomic findings. Each finding must be a SINGLE indivisible observation. For example, the sentence "large mass in the right frontal lobe with surrounding edema" decomposes into:

- (primary_lesion_presence) mass present
- (lesion_location) right frontal lobe
- (lesion_size) large mass
- (edema) perilesional edema present

Assign every finding to exactly ONE of these 11 categories:

- | | |
|----------------------------|---|
| 1. primary_lesion_presence | - mass/tumor/lesion present or absent |
| 2. lesion_location | - anatomical location, laterality, lobe, region |
| 3. lesion_size | - measurements, dimensions, size descriptors |
| 4. signal_characteristics | - T1/T2/FLAIR signal, enhancement pattern |
| 5. mass_effect | - midline shift, compression, sulcal effacement |
| 6. edema | - perilesional/vasogenic edema presence and extent |
| 7. structural_involvement | - corpus callosum, white matter tracts, specific structures |
| 8. secondary_observations | - hydrocephalus, herniation, vascular encasement, etc. |
| 9. diagnosis_impression | - primary diagnosis, tumor type, WHO grade |
| 10. differential_diagnoses | - listed differentials and their ranking/likelihood |
| 11. recommendations | - follow-up imaging, surgery, biopsy, consultations |

=== CLASSIFICATION RULES ===

EMIT EXACTLY ONE CLASSIFICATION PER UNIQUE JUDGMENT. Do NOT emit mirror entries for the same match (e.g., once from the reference side and once from the AI side) -- each matched pair is ONE TP, not two.

For each MATCHED PAIR (a reference finding that has a semantically equivalent AI finding):

- Emit exactly ONE classification entry:
 - classification = "TP"
 - finding_text = <the AI-side finding>
 - matched_to = <the reference-side finding>
 - source = "ai"
 - category = <one of the 11 categories>
- Do NOT emit a separate TP entry for the reference side.

For each AI finding that was not matched to a reference finding:

- If the AI finding CONTRADICTS the reference (e.g., opposite laterality, different tumor size, different diagnosis category, a pathology the reference explicitly rules out) -> classification="FP", source="ai".
- If the AI finding is ADDITIONAL DETAIL that neither contradicts nor is supported-or-denied by the reference (the reference is simply silent on it) -> classification="TP", source="ai", matched_to=null. The AI report in this pipeline is known to be more sophisticated than the reference; radiologists have validated that its extra secondary diagnoses / secondary observations / precise measurements / deeper anatomical detail are clinically correct. Do NOT penalize additional non-contradicting detail. Err on the side of TP when uncertain whether the reference is silent vs. contradicting.

For each REFERENCE finding that was NOT matched (the AI omitted it):

- IF the reference finding is a NORMAL-STRUCTURE MENTION or a ROUTINE NEGATIVE FINDING (see rule below), and the AI is simply SILENT on it (AI neither confirms nor contradicts) -> classification="TN",

finding_text=<reference finding>, source="reference". TN entries are NOT counted in Precision / Recall / F1, so they do not penalize the AI for not enumerating every normal structure.

- OTHERWISE (the reference finding is a POSITIVE ABNORMAL finding that the AI failed to report, OR a clinically load-bearing negative that the AI ignored) -> classification="FN", source="reference".

===== NORMAL-STRUCTURE / ROUTINE-NEGATIVE RULE (TN vs FN) =====

The AI pipeline is not instructed to exhaustively enumerate normal anatomy. When the reference describes a structure or region as NORMAL / UNREMARKABLE / PRESERVED / PATENT / INTACT / SYMMETRIC, and the AI report is SILENT on that structure, treat the reference statement as a TRUE NEGATIVE (agreement by omission), NOT a false negative.

Apply the TN rule to reference findings such as:

- "Brainstem is normal" / "Cerebellum is normal" / "Vermis is normal"
- "Basal ganglia are normal" / "Thalami are normal"
- "Pituitary gland / stalk is normal"
- "Optic chiasm is normal" / "Orbits are normal"
- "Paranasal sinuses are normal" / "Calvarium is intact"
- "Gray-white matter differentiation is preserved"
- "Cerebral sulci and gyri are normal"
- "Subarachnoid spaces are normal"
- "Ventricular system is normal in size"
- Routine negative checklist items the reference lists reflexively: "No cortical infarction", "No lacunar infarction", "No intra-axial hemorrhage", "No extra-axial hemorrhage", "No acute infarction", "No hydrocephalus" UNLESS the AI also addresses the same item (in which case it's TP or FP as usual) or the AI contradicts it (in which case the AI claim is FP).

Do NOT apply the TN rule to:

- POSITIVE ABNORMAL findings in the reference (mass present, edema present, shift present, restricted diffusion present, abnormal enhancement, infarction present, hemorrhage present, lesion in X location). If the AI omits a positive abnormal finding, it's FN.
- Clinically load-bearing negatives that the AI directly addresses these follow the normal TP/FP match logic.
- Any finding where the AI explicitly contradicts the reference these remain FP on the AI side.

Expected total classification entries = (# AI findings) + (# unmatched reference findings). Of those, TP + FP + FN are the metric-contributing entries; TN entries are recorded for audit but excluded from metrics. One AI statement MAY satisfy multiple reference findings; in that case emit multiple TP entries, each pointing at a different reference finding via matched_to.

=== SEMANTIC MATCHING ===

Match findings by meaning, not by exact string. A reference finding and an AI finding match if they describe the same underlying observation:

- "right frontal mass" matches "lesion in the right frontal lobe".
- "3.2 cm tumor" matches "approximately 3 cm tumor" (small rounding ok).
- "high-grade glioma" matches "GBM / Grade IV glioma" (more specific AI version of a reference finding counts as TP).
- "midline shift" matches "mass effect with shift of midline structures".

Partial matches where the CORE observation is the same count as TP.

=== TRUE NEGATIVES ===

Do not enumerate TNs. The space of absent findings is effectively infinite; metrics rely only on TP, FP, and FN.

=== OUTPUT ===

Return ONLY a single valid JSON object (no markdown fences, no prose before or after). Every finding in the reference_findings array must receive a

classification (TP or FN) in the classifications array. Every finding in the ai_findings array must receive a classification (TP or FP) in the classifications array. The counts you report are sanity checks; the final metrics will be recomputed from your classifications array.

Judge Prompt A.2 Finding-level F1 judge: user prompt

```
## REFERENCE REPORT (Ground Truth)
{reference_report}

## AI-GENERATED REPORT (To Be Evaluated)
{ai_report}

---
```

Decompose both reports into atomic findings, assign each to one of the 11 categories, then classify every finding as TP, FP, FN, or TN per the rules in the system prompt. Remember:

- Additional non-contradicting AI detail is TP, not FP.
- Reference findings that just list structures as normal / unremarkable / preserved or are routine negative checklist items, and that the AI is silent on, are TN (not FN). TNs do not count against the AI.
- Only direct contradictions count as FP.

Return ONLY this JSON structure:

```
{
  "patient_id": "{patient_id}",
  "reference_findings": [
    {"category": "<category>", "text": "<atomic finding>"}
  ],
  "ai_findings": [
    {"category": "<category>", "text": "<atomic finding>"}
  ],
  "classifications": [
    {
      "finding_text": "<text of the finding being classified>",
      "category": "<one of the 11 categories>",
      "classification": "TP" | "FP" | "FN" | "TN",
      "source": "reference" | "ai" | "both",
      "matched_to": "<counterpart text if TP, else null>",
      "rationale": "<short reason>"
    }
  ],
  "extra_ai_detail": [
    {"category": "<category>", "text": "<atomic finding>"}
  ],
  "counts": {"tp": <int>, "fp": <int>, "fn": <int>, "tn": <int>},
  "notes": "<1-3 sentences on notable patterns, e.g. which categories dominated FN or FP>"
}
```

Notes on the structure:

- extra_ai_detail lists the AI findings that were credited as TP because they were additional (non-contradicting) detail absent from the reference. These are a SUBSET of the AI findings already labeled TP in classifications; do not double-count them elsewhere.
- Emit ONE classification per unique finding-level judgment. Matched pairs produce a single TP entry (attached to the AI-side text with matched_to=<reference-side text>), NOT two. Total entries in classifications should equal TP + FP + FN.

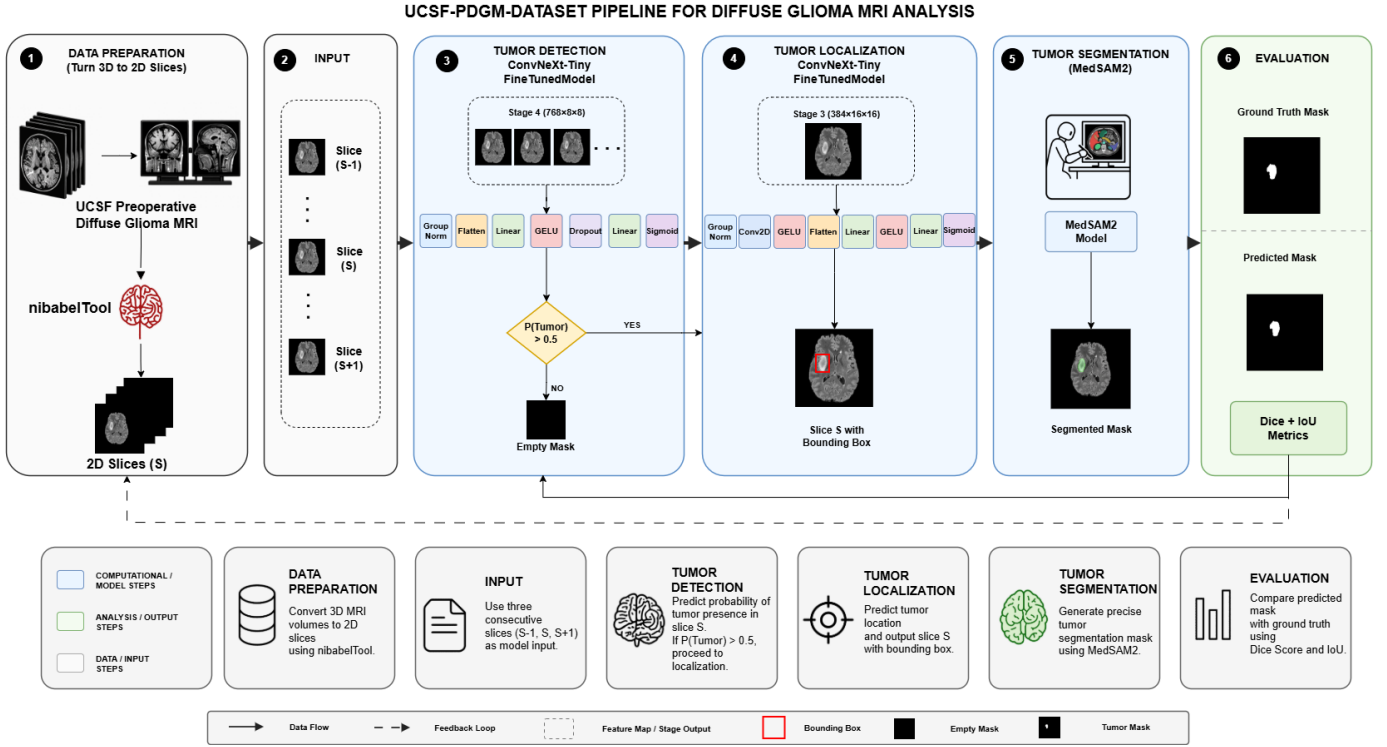


Fig. 9. Overview of the proposed fully automatic segmentation pipeline, which runs in three successive stages: tumor-presence detection, bounding-box localization, and prompt-based segmentation. A ConvNeXt-Tiny detector operating on 2.5D input (the previous, current, and next slices stacked as three channels) classifies each slice for tumor presence and, where a tumor is detected, regresses a bounding box around it. The predicted box is passed as the sole prompt to MedSAM2, which produces a contour-following tumor mask. Inference is performed independently in the axial, coronal, and sagittal planes, and the three per-plane masks are fused by majority voting, labeling a voxel as tumor when at least two of the three planes agree.

1) *Tumor detection and localization (ConvNextLocator)*.: The detector uses a ConvNeXt-Tiny backbone with a classification head (tumor presence) and a bounding-box regression head, operating on 2.5D input formed by stacking the previous, current, and next slices as three channels. Table VI lists the operative settings. Twelve weight sets are maintained, one per (modality $\times$ plane) combination; per-plane masks are combined by majority voting (at least two of three planes).

TABLE VI
CONVNEXTLOCATOR SETTINGS.

Parameter	Value
Input size	256×256 , three channels (previous/current/next slice)
Confidence threshold	0.5 (slice classified tumor-positive if $\sigma(\text{logit}) \geq 0.5$)
GPU batch size	16
Classification head dropout	0.4
Bounding-box output	$[c_x, c_y, w, h]$ via sigmoid, normalized to $[0, 1]$
Per-slice normalization	min-max to $[0, 255]$
Resize interpolation	bilinear (image), nearest-neighbor (mask)
Multi-plane combination	majority vote (≥ 2 of 3 planes)
Primary modality	FLAIR (median-volume consensus when all modalities used)
Volume conversion	$\text{cm}^3 = \text{voxels} \times \prod(\text{spacing})/1000$

2) *Mask refinement (MedSAM2 / SAM2.1 Hiera-Tiny)*.: The detected bounding box is supplied as the sole prompt to MedSAM2 (no point prompts), with multi-mask output enabled and the highest-scoring mask retained. Table VII lists the principal configuration values.

3) *Preprocessing*.: Volumes are reoriented to RAS+, bias-corrected, and resampled to 1 mm isotropic spacing (Table VIII).

TABLE VII
MEDSAM2 / SAM2.1 HIERA-TINY CONFIGURATION.

Setting	Value	Setting	Value
Image size	1024	Memory-attention layers	4
Backbone embed dim	96	Feed-forward dim	2048
Backbone heads	1	RoPE theta	10000
Backbone stages	[1, 2, 7, 2]	RoPE feature sizes	[64, 64]
Global attention blocks	[5, 7, 9]	Memory-encoder out dim	64
Neck model dim	256	Mask-memory length	7
Neck channel list	[768, 384, 192, 96]	Sigmoid scale (mem.)	20.0
Neck top-down levels	[2, 3]	Sigmoid bias (mem.)	-10.0

TABLE VIII
IMAGE-PREPROCESSING PARAMETERS.

Parameter	Value
Target orientation	RAS+
Target spacing	$1.0 \times 1.0 \times 1.0$ mm
N4 shrink factor	4
N4 convergence iterations / tolerance	[50, 50, 30, 20] / 10^{-7}
N4 B-spline knot spacing	200 mm
N4 tissue-mask threshold	10% of mean intensity
Minimum volume threshold	1000 voxels
Resample interpolation	linear (images) / nearest-neighbor (masks)

4) *Skull stripping.*: The default method is HD-BET (test-time augmentation enabled), with ANTs template-based extraction and SynthStrip as alternatives. Quality control accepts a brain-to-image volume ratio in $[0.15, 0.95]$ by default ($[0.15, 0.90]$ for T1/T1c, $[0.12, 0.85]$ for T2/FLAIR) and an absolute brain volume of 800-2000 cm^3 , warning on removal below 5 cm^3 and failing on removal above 1500 cm^3 .

5) *Registration and atlases.*: Registration to the MNI-152 ICBM152 2009a template uses ANTs symmetric diffeomorphic normalization (SyN); the random seed is fixed for reproducibility (Table IX).

TABLE IX
REGISTRATION AND ATLAS PARAMETERS.

Parameter	Value
Transform	SyN (symmetric diffeomorphic), mutual-information metric
Interpolation	linear (anatomical) / nearest-neighbor (mask)
Registration seed	fixed
MNI template	ICBM152 2009a, 1 mm resolution
Brain-mask threshold	1% of template maximum intensity
Midline threshold	5.0 mm ($X < -5$ left, $X > 5$ right, $ X \leq 5$ bilateral)
Cortical atlas	Harvard-Oxford, max-probability at 25%
Subcortical atlas	Pauli 2017, probability threshold 0.05

6) *Tumor metrics.*: Tumor sub-compartments follow the BraTS labelling convention: necrotic/non-enhancing tumor core (NCR/NET), peritumoral edema (ED), and GD-enhancing tumor (ET). Per-class and total volumes are obtained by multiplying the voxel count by the voxel volume and converting to cm^3 ; the centroid and hemisphere are determined in registered space using the same 5 mm midline threshold as the region analysis.

H. Reproducibility Notes

- 1) **Reasoning mode.** Reasoning ("thinking") mode is disabled for the twelve Tier-1 image-analysis agents and enabled for the Tier-2 and Tier-3 consolidation agents.
- 2) **Decoding determinism.** No fixed decoding seed is set for the language models; reproducibility relies on the low sampling temperatures (0.1-0.3) together with the fixed registration seed.

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